

Asymptotics of n -universal lattices over number fields

Quadratic Lattices in Prague, Charles University

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Preliminaries

Setup:

- K is a totally real number field of degree d , discriminant Δ_K , maximal order $\mathcal{O}_K$.
 $\mathcal{O}_K^+$ is the totally positive elements in $\mathcal{O}_K$.
- L will always denote an $\mathcal{O}_K$ -**lattice**.
(i.e. a finitely generated torsion-free $\mathcal{O}_K$ -module, equipped with a quadratic map $Q : L \rightarrow \mathcal{O}_K$ where $Q(a\mathbf{v}) = a^2 Q(\mathbf{v})$ for all $a \in \mathcal{O}_K$, $\mathbf{v} \in L$ and $B(\mathbf{v}, \mathbf{w}) := \frac{1}{2}(Q(\mathbf{v} + \mathbf{w}) - Q(\mathbf{v}) - Q(\mathbf{w}))$ is $\mathcal{O}_K$ -bilinear)
- The **rank** of an $\mathcal{O}_K$ -lattice L is $\dim_K(L \otimes K)$.
- An $\mathcal{O}_K$ -lattice L is **classical** if $B(\mathbf{v}, \mathbf{w}) \in \mathcal{O}_K$ for all $\mathbf{v}, \mathbf{w} \in L$.
- An $\mathcal{O}_K$ -lattice L is **totally positive definite** if $Q(\mathbf{v}) \succ 0$ for all nonzero $\mathbf{v} \in L$.

Throughout this talk, every $\mathcal{O}_K$ -lattice is *classical* and *totally positive definite* (unless otherwise specified).

Universal $\mathcal{O}_K$ -lattices

Universal $\mathcal{O}_K$ -lattice

A (classical, totally positive definite) $\mathcal{O}_K$ -lattice L is **universal** if it represents every $\alpha \in \mathcal{O}_K^+$; i.e. if for every totally positive $\alpha \in \mathcal{O}_K$, there is $\mathbf{v} \in L$ such that $Q(\mathbf{v}) = \alpha$.

Examples:

- I_4 is universal over $\mathbb{Z}$. (Lagrange 1770)
- I_3 is universal over $\mathbb{Z}\left[\frac{1+\sqrt{5}}{2}\right]$. (Maass 1941)
- $\langle 1, 1, 2 + \sqrt{2} \rangle$ is universal over $\mathbb{Z}[\sqrt{2}]$. (Chan–Kim–Raghavan 1996)
- $\begin{pmatrix} 1 & 1 \\ 1 & \frac{5+\sqrt{13}}{2} \end{pmatrix} \perp \langle 1, \frac{5-\sqrt{13}}{2} \rangle$ is universal over $\mathbb{Z}\left[\frac{1+\sqrt{13}}{2}\right]$. (Sasaki 2009)
- $I_4 \perp \gamma I_4$, with $\gamma = D + \sqrt{D^2 - 1}$, is universal over $\mathbb{Z}[\sqrt{D^2 - 1}]$, for any squarefree $D^2 - 1$. (Kim 2000)

How small can a universal $\mathcal{O}_K$ -lattice be?

Minimal ranks

We write $m_*(\mathcal{O}_K)$ for the smallest rank of a universal $(*)$ $\mathcal{O}_K$ -lattice L .

Here the subscript $*$ records what conditions L must satisfy:

$m_{\text{nc}}(\mathcal{O}_K)$	not necessarily classical (i.e. no condition)
$m_{\text{cl}}(\mathcal{O}_K)$	classical
$m_{\text{cl,free}}(\mathcal{O}_K)$	classical and free (i.e. classical quadratic form)
$m_{\text{diag}}(\mathcal{O}_K)$	diagonal

For each totally real K : $3 \leq m_{\text{nc}}(\mathcal{O}_K) \leq m_{\text{cl}}(\mathcal{O}_K) \leq m_{\text{cl,free}}(\mathcal{O}_K) \leq m_{\text{diag}}(\mathcal{O}_K)$.

Conjecture (Kitaoka 1990s)

There are only finitely many totally real fields K such that $m_{\text{cl}}(\mathcal{O}_K) = 3$.

Minimal ranks: the story so far

A (non-exhaustive) summary of some progress towards Kitaoka's conjecture:

- (Chan–Kim–Raghavan 1996) For a real quadratic $K = \mathbb{Q}(\sqrt{D})$, D squarefree, $m_{\text{cl,free}}(\mathcal{O}_K) = 3$ if and only if $D \in \{2, 3, 5\}$.
- (Earnest–Khosravani 1997) If K has odd degree, then $m_{\text{nc}}(\mathcal{O}_K) \geq 4$.
- (Kim 1999) There are only finitely many real quadratic K with $m_{\text{diag}}(\mathcal{O}_K) \leq 7$.
- (Kim 2000) There are infinitely many real quadratic K with $m_{\text{diag}}(\mathcal{O}_K) = 8$.
- (Blomer–Kala 2015, Kala 2016) For any $M \geq 1$, there exist infinitely many real quadratic K with $m_{\text{nc}}(\mathcal{O}_K) \geq M$.
- (Yatsyna 2019) For any $M \geq 1$, there exist infinitely many totally real cubic K with $m_{\text{nc}}(\mathcal{O}_K) \geq M$.
- (Kala–Svoboda 2019) For any fixed $k \geq 1$ and $M \geq 1$, there are infinitely many degree 2^k multiquadratic K with $m_{\text{nc}}(\mathcal{O}_K) \geq M$.
- (Krásenský–Tinková–Zemková 2020) For all totally real biquadratic K , $m_{\text{cl,free}}(\mathcal{O}_K) \geq 4$.

Minimal ranks: the story so far

- (Kim 2003, Kala–Yatsyna 2021) For any fixed $d \geq 1$, there are only finitely many totally real degree d fields K with $m_{\text{nc}}(\mathcal{O}_K) \leq 3$.
- (Kim–Kim–Park 2022) There are only finitely many real quadratic K with $m_{\text{nc}}(\mathcal{O}_K) \leq 7$.
- (Kala 2023) For any fixed $d \geq 2$ divisible by either 2 or 3, and $M \geq 1$, there exist infinitely many degree d fields K such that $m_{\text{nc}}(\mathcal{O}_K) \geq M$.
- (Kala–Yatsyna–Žmija 2023) For any $M \geq 1$, the set of real quadratic K such that $m_{\text{nc}}(\mathcal{O}_K) \leq M$ has density 0.
- (Man 2024). For any fixed $k \geq 1$ and $M \geq 1$, the set of degree 2^k multiquadratic fields K such that $m_{\text{nc}}(\mathcal{O}_K) \leq M$ has density 0.
- (Kala–Krásenský–Park–Yatsyna–Žmija 2025) If $K = \mathbb{Q}(\sqrt{D})$ is real quadratic, D squarefree, and $m_{\text{nc}}(\mathcal{O}_K) \leq 3$, then $D \in \{2, 3, 5, 6, 7, 10, 13, 17, 21, 33, 41, 65, 77\}$.
- (Kala–Kramer–Krásenský 2025) If Δ_K odd, then $m_{\text{cl,free}}(\mathcal{O}_K) = 3 \iff K = \mathbb{Q}(\sqrt{5})$
- (Kramer–Krásenský 2026) For all totally real quartic fields K , $m_{\text{cl,free}}(\mathcal{O}_K) \geq 4$.

n -universal $\mathcal{O}_K$ -lattices

n -universal quadratic lattices

A (classical, totally positive definite) $\mathcal{O}_K$ -lattice L is **n -universal** if it represents every $\mathcal{O}_K$ -lattice of rank n (i.e. if for every totally positive rank n $\mathcal{O}_K$ -lattice M , there is an $\mathcal{O}_K$ -linear $\sigma : M \rightarrow L$ with $Q_M = Q_L \circ \sigma$).

Examples:

- I_5 is 2-universal over $\mathbb{Z}$. (Mordell 1930)
- I_8 is 5-universal over $\mathbb{Z}$. (Ko 1937)
- $I_3 \perp \begin{pmatrix} 2 + \sqrt{2} & 1 \\ 1 & 2 - \sqrt{2} \end{pmatrix} \perp \langle 2 + \sqrt{2} \rangle$ is 2-universal over $\mathbb{Z}[\sqrt{2}]$. (Sasaki 2006)
- $I_4 \perp \begin{pmatrix} 2 & \frac{1+\sqrt{5}}{2} \\ \frac{1+\sqrt{5}}{2} & 2 \end{pmatrix}$ is 2-universal over $\mathbb{Z}[\frac{1+\sqrt{5}}{2}]$. (Sasaki 2006)

We write $U_{\mathcal{O}_K}(n)$ for the minimal rank of an n -universal $\mathcal{O}_K$ -lattice.

n -universal lattices over $\mathbb{Z}$

The values for $U_{\mathbb{Z}}(n)$ have been computed for all $n \leq 10$:

n	$U_{\mathbb{Z}}(n)$	Authors	Year
1	4	J.-L. Lagrange, ...	1770+
2	5	L. J. Mordell	1930
3	6	C. Ko	1937
4	7	C. Ko	1937
5	8	C. Ko	1937

n	$U_{\mathbb{Z}}(n)$	Authors	Year
6	13	B.-K. Oh	2000
7	15	B.-K. Oh	2000
8	16	B.-K. Oh	2000
9	28	B.-K. Oh	2000
10	30	B.-K. Oh	2000

Theorem (J. Yoon 2026+)

$$42 \leq U_{\mathbb{Z}}(11) \leq 45.$$

n -universal lattices over $\mathbb{Z}$

By counting indecomposable lattices, we can derive the following bounds for $U_{\mathbb{Z}}(n)$:

$32 \leq U_{\mathbb{Z}}(12) \leq 64$	$728 \leq U_{\mathbb{Z}}(21) \leq 6\,673$
$33 \leq U_{\mathbb{Z}}(13) \leq 105$	$1\,323 \leq U_{\mathbb{Z}}(22) \leq 15\,849$
$48 \leq U_{\mathbb{Z}}(14) \leq 123$	$2\,428 \leq U_{\mathbb{Z}}(23) \leq 65\,146$
$64 \leq U_{\mathbb{Z}}(15) \leq 196$	$6\,673 \leq U_{\mathbb{Z}}(24) \leq 456\,350$
$105 \leq U_{\mathbb{Z}}(16) \leq 254$	$15\,849 \leq U_{\mathbb{Z}}(25) \leq 10\,452\,029$
$123 \leq U_{\mathbb{Z}}(17) \leq 475$	$65\,146 \leq U_{\mathbb{Z}}(26) \leq \dots$
$196 \leq U_{\mathbb{Z}}(18) \leq 728$	$456\,350 \leq U_{\mathbb{Z}}(27) \leq \dots$
$254 \leq U_{\mathbb{Z}}(19) \leq 1\,323$	$10\,452\,029 \leq U_{\mathbb{Z}}(28) \leq \dots$
$475 \leq U_{\mathbb{Z}}(20) \leq 2\,428$	$1\,099\,806\,087 \leq U_{\mathbb{Z}}(29) \leq \dots$

We want to show that $U_{\mathbb{Z}}(n)$ grows very quickly as $n \rightarrow \infty$.

Asymptotics for $U_{\mathcal{O}_K}(n)$

Theorem (Park–V–Yatsyna–Yoon 2025)

Let K be a totally real degree d number field of discriminant Δ_K . Then, with the exception of a finite set of at most 559 real quadratic fields K , we have

$$\log U_{\mathcal{O}_K}(n) = \frac{d}{4}n^2 \log n + n^2 \left(\frac{2 \log \Delta_K - 2d \log(2\pi) - 3d}{8} \right) + \mathcal{O}(n \log n)$$

as $n \rightarrow \infty$.

- In particular, $U_{\mathcal{O}_K}(n)$ grows very quickly: $U_{\mathcal{O}_K}(n) > n^{dn^2/5}$ for all sufficiently large n .
- The set of exceptional real quadratic fields are $\mathbb{Q}(\sqrt{D})$ for all squarefree $D \leq 917$ and $D = 929$.
- In fact the $\leq$ holds for all totally real fields K , and the $\geq$ holds for infinitely many n .

Proof strategy

Our proof strategy is to obtain bounds for $U_{\mathcal{O}_K}(n)$ through a series of reductions:

1. Use Siegel's mass formula to bound $\text{mass}_K(L)$ for rank n unimodular $\mathcal{O}_K$ -lattices L .
2. Use bounds on the maximal orders of finite subgroups of $\text{GL}_n(K)$ to obtain bounds on the class numbers $\text{cl}_K(L)$.
3. Hence obtain bounds on $b_{\mathcal{O}_K}(m)$: the number of rank m unimodular $\mathcal{O}_K$ -lattices.
4. Apply a combinatorial theorem of Wright to bound $a_{\mathcal{O}_K}(m)$: the number of *indecomposable* rank m unimodular $\mathcal{O}_K$ -lattices.
5. Hence obtain bounds for $U_{\mathcal{O}_K}(n)$: the minimal rank of an n -universal $\mathcal{O}_K$ -lattice.

$$\text{mass}_K(L) \longrightarrow \text{cl}_K(L) \longrightarrow b_{\mathcal{O}_K}(m) \longrightarrow a_{\mathcal{O}_K}(m) \longrightarrow U_{\mathcal{O}_K}(n)$$

Step 1: Siegel's mass formula

- For any $\mathcal{O}_K$ -lattice L , Siegel defined the following genus-invariant constant:

$$\text{mass}_K(L) := \sum_{\Lambda \in \text{gen}(L)} \frac{1}{\#\text{Aut}(\Lambda)}$$

where $\text{gen}(L)$ denotes the (finitely many) isometry classes of $\mathcal{O}_K$ -lattices on $L \otimes K$ that are everywhere locally isometric to L .

- Since L is positive definite, $\text{Aut}(\Lambda)$ is finite and contains ± 1 , thus we have the trivial bounds:

$$2 \text{mass}_K(L) \leq \text{cl}_K(L) \leq \text{mass}_K(L) \cdot \max_{\Lambda \in \text{gen}(L)} \#\text{Aut}(\Lambda).$$

Step 1: Siegel's mass formula

Theorem (Siegel's mass formula (1935, 1937))

Let L be a positive definite unimodular $\mathcal{O}_K$ -lattice of rank $n \geq 2$. Then

$$\text{mass}_K(L) = \Delta_K^{n(n-1)/4} \cdot \frac{1}{\alpha_\infty(L)} \cdot \prod_{\mathfrak{p}} \frac{1}{\alpha_{\mathfrak{p}}(L)},$$

the product running over all finite places $\mathfrak{p}$ of K .

Here, $\alpha_\infty(L)$ and $\alpha_{\mathfrak{p}}(L)$ are the local densities:

$$\alpha_\infty(L) := \left(\frac{\pi^{n(n+1)/4}}{2\Gamma(\frac{1}{2})\Gamma(\frac{2}{2}) \cdots \Gamma(\frac{n}{2})} \right)^d,$$
$$\alpha_{\mathfrak{p}}(L) := \lim_{\nu \rightarrow \infty} \frac{\#\{X \in \text{Mat}_n(\mathcal{O}_K/\mathfrak{p}^\nu) : X^T G_{\mathfrak{p}} X \equiv G_{\mathfrak{p}} \pmod{\mathfrak{p}^\nu}\}}{2N(\mathfrak{p})^{\nu n(n-1)/2}},$$

where $G_{\mathfrak{p}}$ is a Gram matrix of the localisation $L_{\mathfrak{p}}$.

The mass of I_n

Theorem (Körner 1973)

Let K be a totally real field and let $n \geq 3$ be an odd integer. For each dyadic prime $\mathfrak{p}$ in K , let $e_{\mathfrak{p}}$ and $f_{\mathfrak{p}}$ be the ramification index and residue degree of $\mathfrak{p}$ respectively. Let $\xi(n, \mathfrak{p})$ be defined as

$$\frac{\xi(n, \mathfrak{p})}{N(\mathfrak{p})^{(1-n)\lfloor e_{\mathfrak{p}}/2 \rfloor - e_{\mathfrak{p}}}} = \begin{cases} 1 & \text{if } 2 \mid e_{\mathfrak{p}}, \\ \frac{1}{2} \left(1 + \frac{(-1)^{f_{\mathfrak{p}} \lfloor (n+1)/4 \rfloor}}{N(\mathfrak{p})^{\lfloor (n-1)/2 \rfloor}} \right) & \text{if } 2 \nmid e_{\mathfrak{p}}. \end{cases}$$

Then

$$\text{mass}_K(I_n) = \left(\frac{2\Gamma(\frac{1}{2})\Gamma(\frac{2}{2}) \cdots \Gamma(\frac{n}{2})}{\pi^{n(n+1)/4}} \right)^{[K:\mathbb{Q}]} \Delta_K^{n(n-1)/4} \prod_{\mathfrak{p}|2} \xi(n, \mathfrak{p}) \prod_{i=1}^{\lfloor (n-1)/2 \rfloor} \zeta_K(2i),$$

where $\Gamma(s)$ is the usual Gamma function, $\zeta_K(s)$ is the Dedekind zeta function of K .

Asymptotics for $\text{mass}_K(I_n)$

Theorem

Let K be a totally real degree d field, and for each dyadic prime $\mathfrak{p}$ in K , let $e_{\mathfrak{p}}$ denote the ramification degree of $\mathfrak{p}$. Then

$$\text{mass}_K(I_n) \sim C_K \left(\frac{n}{2\pi e \sqrt{e}} \right)^{dn^2/4} \left(\frac{8\pi e}{n} \right)^{dn/4} \left(\frac{1}{n} \right)^{d/24} \Delta_K^{n(n-1)/4} \left(\prod_{\mathfrak{p}|2} \frac{1}{N(\mathfrak{p})^{\lfloor e_{\mathfrak{p}}/2 \rfloor}} \right)^n$$

as $n \rightarrow \infty$. Here C_K is the explicit constant

$$C_K = 2^{-5d/4} e^{d/24} \exp \left(\frac{1}{12} - \zeta'(-1) \right)^{-d/2} \cdot \left(\prod_{\mathfrak{p}|2} \frac{N(\mathfrak{p})^{\lfloor e_{\mathfrak{p}}/2 \rfloor - e_{\mathfrak{p}}}}{2^{\delta_{2 \nmid e_{\mathfrak{p}}}}} \right) \cdot \prod_{i=1}^{\infty} \zeta_K(2i)$$

where $\delta_{2 \nmid e_{\mathfrak{p}}}$ is 1 if $e_{\mathfrak{p}}$ is odd, and 0 otherwise.

Step 2: Bounding automorphism groups

Note that $\text{Aut}(L)$ is a finite subgroup of $\text{GL}(L \otimes K) \cong \text{GL}_n(K)$. We can use a classical theorem of Schur:

Theorem (Schur 1905, odd ℓ)

Let G be a finite ℓ -subgroup of $\text{GL}_n(\mathbb{C})$ such that $\text{Tr}(g) \in K$ for all $g \in G$. Then:

$$v_\ell(\#G) \leq m_\ell \left\lfloor \frac{n}{t_\ell} \right\rfloor + \left\lfloor \frac{n}{\ell t_\ell} \right\rfloor + \left\lfloor \frac{n}{\ell^2 t_\ell} \right\rfloor + \dots$$

where $t_\ell := [K(\zeta_\ell) : K]$ and m_ℓ is the maximal integer a such that $K(\zeta_\ell)$ contains ζ_{ℓ^a} .

Corollary

Let L be a positive definite $\mathcal{O}_K$ -lattice. Then $\log \# \text{Aut}(L) \leq n \log n + O_K(n)$.

Step 3: Counting rank n unimodular $\mathcal{O}_K$ -lattices

Let $b_{\mathcal{O}_K}(n)$ be the number of isometry classes of rank n unimodular $\mathcal{O}_K$ -lattices. Note:

$$2\text{mass}_K(I_n) \leq \text{cl}_K(I_n) \leq b_{\mathcal{O}_K}(n) = \sum_{L \in \mathcal{G}_n} \text{cl}_K(L) \leq \#\mathcal{G}_n \cdot \max_L (\text{mass}_K(L) \cdot \#\text{Aut } L)$$

where $\mathcal{G}_n$ is the set of genera of rank n unimodular $\mathcal{O}_K$ -lattices. Applying the known bounds for the $\text{mass}_K(L)$, $\#\text{Aut } L$, and the upper bound $\#\mathcal{G}_n \leq (4\sqrt{5})^d$, we obtain the following theorem:

Theorem

For any totally real field K ,

$$\log b_{\mathcal{O}_K}(n) = \frac{d}{4}n^2 \log n + n^2 \left(\frac{2 \log \Delta_K - 2d \log(2\pi) - 3d}{8} \right) + \mathcal{O}(n \log n)$$

as $n \rightarrow \infty$.

Step 4: Counting indecomp. unimodular lattices

Let $a_{\mathcal{O}_K}(n)$ be the number of isometry classes of *indecomposable* rank n unimodular $\mathcal{O}_K$ -lattices. We have that $a_{\mathcal{O}_K}(n)$ is the *inverse Euler transform* of $b_{\mathcal{O}_K}(n)$:

$$\prod_{m=1}^{\infty} (1 - X^m)^{-a_{\mathcal{O}_K}(m)} = 1 + \sum_{m=1}^{\infty} b_{\mathcal{O}_K}(m) X^m.$$

Theorem (Wright 1968)

Let $\{T_n\}_{n \geq 1}$ be the Euler transform $\{t_n\}_{n \geq 1}$. If $\sum_{s=1}^{n-1} H_s H_{n-s} = o(H_n)$ as $n \rightarrow \infty$, where $\{H_n\}_{n \geq 1}$ is either $\{T_n\}$ or $\{t_n\}$, then $T_n \sim t_n$ as $n \rightarrow \infty$.

- We verified that Wright's theorem holds for $H_n = b_{\mathcal{O}_K}(n)$ over all totally real fields K , with the exceptional of 559 real quadratic fields K .

Step 5: Bounds for $U_{\mathcal{O}_K}(n)$

Lemma

Let K be totally real field. Let $\overline{\mathcal{I}}(m)$ denote all indecomposable unimodular $\mathcal{O}_K$ -lattices of rank at most m . Then

$$\sum_{L \in \overline{\mathcal{I}}(n)} \left\lfloor \frac{n}{\text{rank } L} \right\rfloor (\text{rank } L) \leq U_{\mathcal{O}_K}(n) \leq \sum_{L \in \overline{\mathcal{I}}(n+3)} \left\lfloor \frac{n+3}{\text{rank } L} \right\rfloor \cdot (\text{rank } L).$$

Sketch of lower bound:

- Use uniqueness of orthogonal decomposition to show that any n -universal $\mathcal{O}_K$ -lattice M represents

$$\bigperp_{L \in \overline{\mathcal{I}}(n)} \left\lfloor \frac{n}{\text{rank } L} \right\rfloor L.$$

Sketch of upper bound:

- Prove that M represents every unimodular lattice on $I_{n+3} \otimes K$, where M is

$$\bigperp_{L \in \overline{\mathcal{I}}(n+3)} \left\lfloor \frac{n+3}{\text{rank } L} \right\rfloor L$$

Step 5: Bounds for $U_{\mathcal{O}_K}(n)$

By the previous lemma, we have

$$\log a_{\mathcal{O}_K}(n) \leq \log U_{\mathcal{O}_K}(n) \leq \log b_{\mathcal{O}_K}(n+3) + 2 \log(n+3).$$

Applying the upper bound obtained for $b_{\mathcal{O}_K}(n)$ in Step 3, and the lower bound obtained for $a_{\mathcal{O}_K}(n)$ in Step 4, this proves the claimed asymptotic for $U_{\mathcal{O}_K}(n)$, so we are done! $\square$

Pfeuffer's results.

By using the Siegel mass formula, Pfeuffer proved the following two finiteness theorems for class numbers of lattices L :

Theorem (Pfeuffer 1979)

Let $C > 0$. Then there are only finitely many totally real fields K such that there exists an $\mathcal{O}_K$ -lattice L of rank ≥ 3 with $\text{cl}_K(L) < C$. Each such field K admits only a finite number of lattices L with bounded scale.

Theorem (Pfeuffer 1979)

Let $C > 0$ and fix a positive integer d . Then there are only finitely many totally real fields K of degree d such that there exists an $\mathcal{O}_K$ -lattice L of rank 2 with $\text{cl}_K(L) < C$. Only finitely many of all these fields (of arbitrary degree) can be reached by a tower of relatively normal extensions from $\mathbb{Q}$. There are finitely many such fields K in all, provided either the generalized Riemann hypothesis or the Artin conjecture (on the analyticity of Artin L -functions) is true.

Extending Pfeuffer to minimal ranks

Theorem (Park–V–Yatsyna–Yoon 2025)

Let $C > 0$ and fix some positive integer $n \geq 3$. Then there are only finitely many totally real fields K such that $U_{\mathcal{O}_K}(n) < C$. In particular, all such fields K satisfying $U_{\mathcal{O}_K}(n) < C$ are contained within an effectively computable finite set of fields.

If $n \geq 3$ and $U_{\mathcal{O}_K}(n) < C$, then the degree d and discriminant Δ_K of such fields K satisfy the following bounds:

$$d \leq \frac{4n \log(C) + 8 \log(n!) + 254n(n-1) - 4 \log(2)}{\log((2 \cdot 60.1^{n(n-1)/4} \prod_{i=1}^n \Gamma(i/2)) / (2^{(n+5)/2} \cdot \pi^{n(n+1)/4} \cdot (2\zeta(n/2))^{\delta_{2|n}}))},$$
$$\Delta_K \leq \left(\frac{C^n (n!)^2 \pi^{dn(n+1)/4} 2^{d(n+5)/2} (2\zeta(n/2))^{d\delta_{2|n}}}{2 \cdot (2 \prod_{i=1}^n \Gamma(i/2))^d} \right)^{4/(n(n-1))},$$

where $\delta_{2|n} = 1$ if n is even, and 0 otherwise.

Case $n = 2$

Theorem (Park–V–Yatsyna–Yoon 2025)

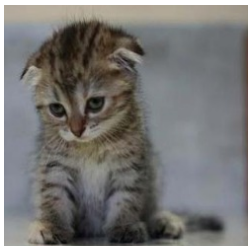
Let $C > 0$, and fix some positive integer $d \geq 1$. Then there are only finitely many degree d totally real fields K such that $U_{\mathcal{O}_K}(2) < C$. In particular, if $d \geq 3$, then all such fields K satisfying $U_{\mathcal{O}_K}(2) < C$ are contained within an effectively computable finite set of fields.

If $d \geq 3$ and $U_{\mathcal{O}_K}(2) < C$, the discriminant Δ_K of such fields K satisfies

$$\Delta_K \leq \max \left(\exp((12(d-1))^2), \left(\frac{2C^2 \cdot (\pi \cdot 2^{5/2})^d \cdot d \cdot (2d)!}{0.000181 \cdot ((d-1)/e)^{d-1}} \right)^{\frac{12d}{5d-12}} \right).$$

Moreover, there are only finitely many such fields K in total (of arbitrary degree) which have a normal tower from $\mathbb{Q}$. There are only finitely many such fields K in total, if either the generalised Riemann hypothesis or the Artin conjecture (on the analyticity of Artin L -functions) is true.

So... what have we learnt?



Not being
able to
solve Kitaoka's
conjecture



Being able to
solve the
n-universal analogue
of Kitaoka's
conjecture for all $n > 2$

Criterion sets

Criterion sets

A **criterion set** over $\mathcal{O}_K$ is a set $\mathcal{S} \subset \mathcal{O}_K^+$ such that an $\mathcal{O}_K$ -lattice L is universal if and only if L represents every element of $\mathcal{S}$.

Examples:

- A classical $\mathbb{Z}$ -lattice is universal if and only if it represents every element of $\mathcal{S} = \{1, 2, 3, 5, 6, 7, 10, 14, 15\}$. (Conway-Schneeberger 1993; Bhargava 2000)
- A (possibly non-classical) $\mathbb{Z}$ -lattice is universal if and only if it represents every element of $\mathcal{S} = \{1, 2, 3, 5, 6, 7, 10, 13, 14, 15, 17, 19, 21, 22, 23, 26, 29, 30, 31, 34, 35, 37, 42, 58, 93, 110, 145, 203, 290\}$. (Bhargava-Hanke 2011)
- A classical $\mathbb{Z}[\frac{1+\sqrt{5}}{2}]$ -lattice is universal if and only if it represents every element of $\mathcal{S} = \{1, 2, \frac{5+\sqrt{5}}{2}, \frac{7+\sqrt{5}}{2}, \frac{7-\sqrt{5}}{2}, \frac{10+2\sqrt{5}}{2}, \frac{15+3\sqrt{5}}{2}\}$. (Lee 2008)

n -universal criterion sets

n -universal criterion sets

An n -**universal criterion set** is a set $\mathcal{S}$ of rank n $\mathcal{O}_K$ -lattices such that an $\mathcal{O}_K$ -lattice L is n -universal if and only if L represents every lattice in $\mathcal{S}$.

Examples:

- (Kim–Kim–Oh 1999) A $\mathbb{Z}$ -lattice is 2-universal if and only if it represents:

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}, \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}, \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}, \begin{pmatrix} 2 & 1 \\ 1 & 4 \end{pmatrix}.$$

- (Oh 1999) A $\mathbb{Z}$ -lattice is 8-universal if and only if it represents both I_8 and E_8 (the unique positive-definite even unimodular lattice of rank 8).

Criterion set theorem

Theorem (Park–V–Yatsyna–Yoon 2025)

Let $C > 0$ and let $n \geq 3$. Then there are only finitely many totally real fields K which admit an n -universal criterion set $\mathcal{S}_K(n)$ such that $\#\mathcal{S}_K(n) < C$.

- In particular, if $\#\mathcal{S}_K(n) < C$ then the degree d and discriminant Δ_K of K satisfy the same upper bounds as given in the previous slide.

Theorem (Park–V–Yatsyna–Yoon 2025)

Furthermore, for any fixed $d \geq 1$ there are only finitely many degree d totally real fields K which admit a 2-universal criterion set $\mathcal{S}_K(2)$ such that $\#\mathcal{S}_K(2) < C$.

- In particular, if $d \geq 3$ and $\#\mathcal{S}_K(2) < C$, then the discriminant Δ_K of K satisfies the same upper bound as given in the previous slide.

Further problems

- Can one prove that $\log U_{\mathcal{O}_K}(n) \sim \frac{d}{4} n^2 \log n$ as $n \rightarrow \infty$ for the remaining 559 exceptional real quadratic fields K ?
- For any fixed totally real field K , can we derive an explicit asymptotic for $U_{\mathcal{O}_K}(n)$ itself (and not just $\log U_{\mathcal{O}_K}(n)$) as $n \rightarrow \infty$?
- Can we derive bounds for $U_{\mathcal{O}_K}(1)$?
- Can $U_{\mathcal{O}_K}(n+1) - U_{\mathcal{O}_K}(n)$ be arbitrarily large for any fixed $n \geq 1$?
- Furthermore, given any constant $C > 0$ and $n \geq 1$, are there always only finitely many totally real fields K such that $U_{\mathcal{O}_K}(n+1) - U_{\mathcal{O}_K}(n) < C$? If so, can all such fields be effectively determined?
- For any fixed totally real field K , can we construct explicit criterion sets $\mathcal{S}_K(n)$ for any $\mathcal{O}_K$ -lattice to be n -universal?

Thank you!